

Individual Assignment 5

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1. Set Up

Packages

```
# general use
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# making plots
library(patchwork)
library(NatParksPalettes)
```

Data

```

# taxonomic info
taxon_list <- read_csv(here("data","taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects"
)

# site info
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+")
)

```

2. Cleaning

sites object

```

# creating new object
ncos_sites <- sites |>
  # filtering to only include ncos quarterly sampling sites
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

clean taxon_list

```

taxon_list_clean <- taxon_list |>
  # convert taxon names to snakecase
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))

```

clean water_quality

```
water_quality_clean <- water_quality |>
  clean_names() |>
  semi_join(ncos_sites, by = "site") |>
  unite("date_site", date, site, remove = FALSE) |>
  group_by(date, site, date_site) |>
  summarize(
    med_ph = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  ungroup() |>
  filter(date_site != "2022-08-12_NVBR")
```

clean aq_ins

```
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(
    date, sample_type, site,
    ostracod,
    copepod,
    hemiptera_corixidae_boatman,
    cladocera,
    nematode,
    diptera_ceratopogonidae,
    annelida_oligochaete,
    diptera_chironomid,
    annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(
    cols = ostracod:annelida_polychaete,
    names_to = "taxon",
    values_to = "abundance") |>
```

```

# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5
  ↳ liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(
  select(water_quality_clean,
        date_site,
        med_ph,
        med_sal,
        med_do),
  by = "date_site"
) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
  ↳ = TRUE),
  site_full = fct_rev(site_full))

```

clean aq_ins_clean

```

# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>

```

```
# filter to only include copepods
filter(taxon == "copepod") |>
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))
```

Problems

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: site
- y-axis: ostracod abundance
- geometry to represent average abundance/L: `geom_jitter`
- geometry to represent medians: `stat_summary()`

b. Wrangle your data

```
# create new object from aq_ins_clean
ost_abundance <- aq_ins_clean |>
# filter to only ostracods
filter(taxon == "ostracod") |>
# remove missing abundance/L values
filter(!is.na(ave_lit)) |>
# group by site
group_by(site_full) |>
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1)) |>
# ungroup data frame
ungroup()

slice_sample(ost_abundance, n = 10)
```

A tibble: 10 x 24

	date_site	date	site	taxon	ave_lit	med_ph	med_sal	med_do
	<chr>	<dtm>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	2022-08-12_NEC	2022-08-12 00:00:00	NEC	ostr~	0	7.37	NA	NA
2	2024-01-28_NPB	2024-01-28 00:00:00	NPB	ostr~	3.33	7.43	3.08	6.6

```

3 2023-11-14_NPB 2023-11-14 00:00:00 NPB ostr~ 18 NA NA NA
4 2022-02-23_NVBR 2022-02-23 00:00:00 NVBR ostr~ 0 8.86 45.2 10.2
5 2022-05-12_NPB 2022-05-12 00:00:00 NPB ostr~ 1.07 7.97 11.4 8.17
6 2023-02-12_NEC 2023-02-12 00:00:00 NEC ostr~ 2.13 NA NA NA
7 2023-07-17_NPB 2023-07-17 00:00:00 NPB ostr~ 4.53 NA NA NA
8 2022-08-12_NVBR 2022-08-12 00:00:00 NVBR ostr~ 0 NA NA NA
9 2021-11-23_NPB2 2021-11-23 00:00:00 NPB2 ostr~ 0.267 7.59 1.99 0.65
10 2023-02-21_NEC 2023-02-21 00:00:00 NEC ostr~ 0 8.56 NA 8.8
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>

```

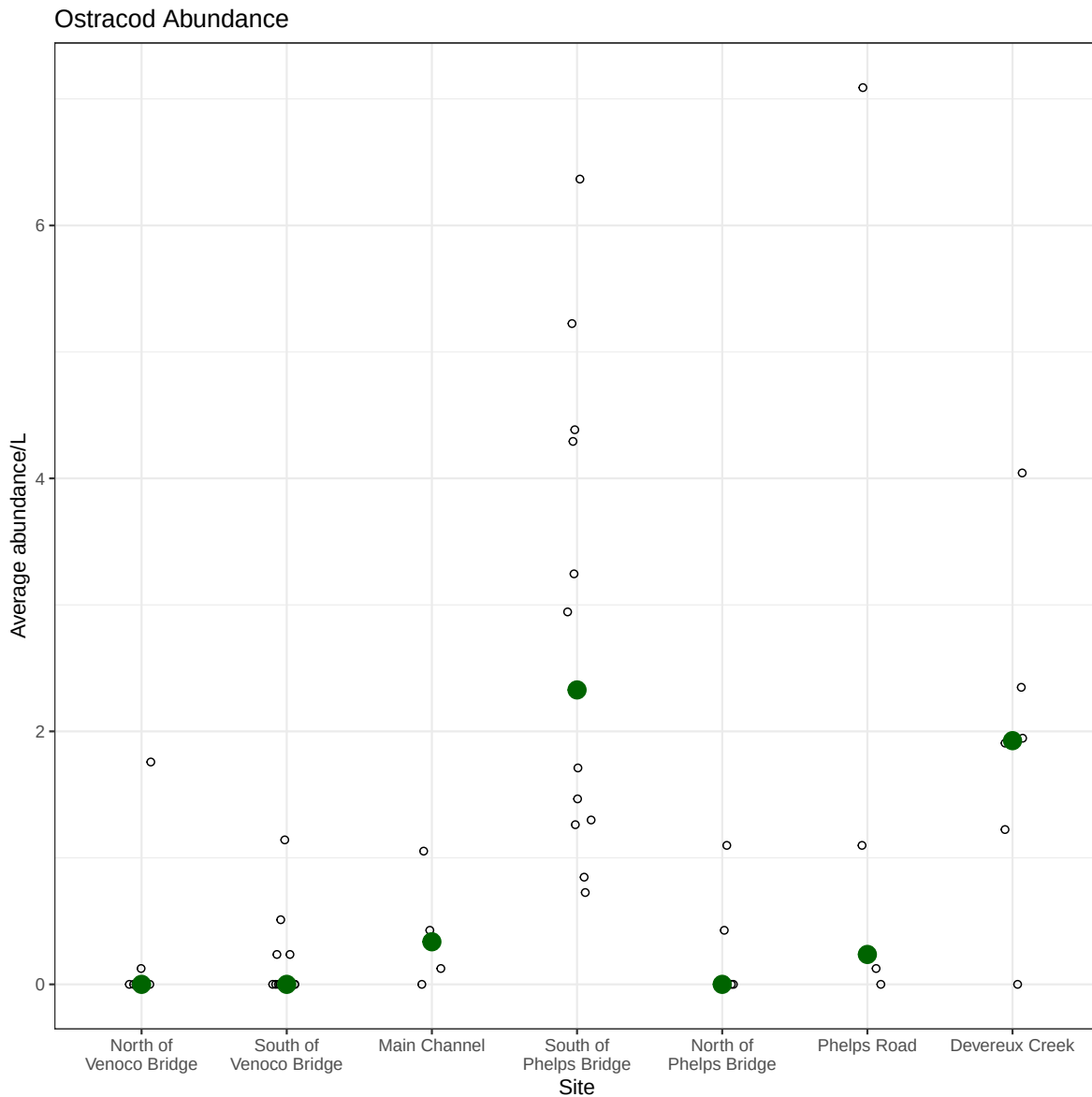
c. Code your figure

```

# base layer
ostracods_plot <- ggplot(data = ost_abundance,
  # x-axis: site
  # y-axis: mean abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
    shape = 21,
    width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
    fun = "median",
    size = 4,
    color = "darkgreen") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
    y = "Average abundance/L",
    title = "Ostracod Abundance") +
  # different theme
  theme_bw()

ostracods_plot

```



d. Create a multipanel figure

```
# base layer
salinity_plot <- ggplot(data = aq_ins_clean,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
```

```

        y = med_sal)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent means
stat_summary(fun = median,
            color = "red",
            size = 1) +
# relabelling x- and y-axes
labs(x = "Site",
     y = "Median salinity (ppt)",
     title = "Site Salinity") +
# different theme
theme_bw()

# base layer
copepods_plot <- ggplot(data = copepods,
                       # x-axis: site
                       # y-axis: mean abundance/L
                       mapping = aes(x = site_full,
                                     y = log_ave_lit)) +
# first layer: points to represent observations
geom_jitter(height = 0,
            shape = 21,
            width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
            fun = "median",
            size = 4,
            color = "blue") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "Average abundance/L",
     title = "Copepod Abundance") +
# different theme
theme_bw()

# base layer

```

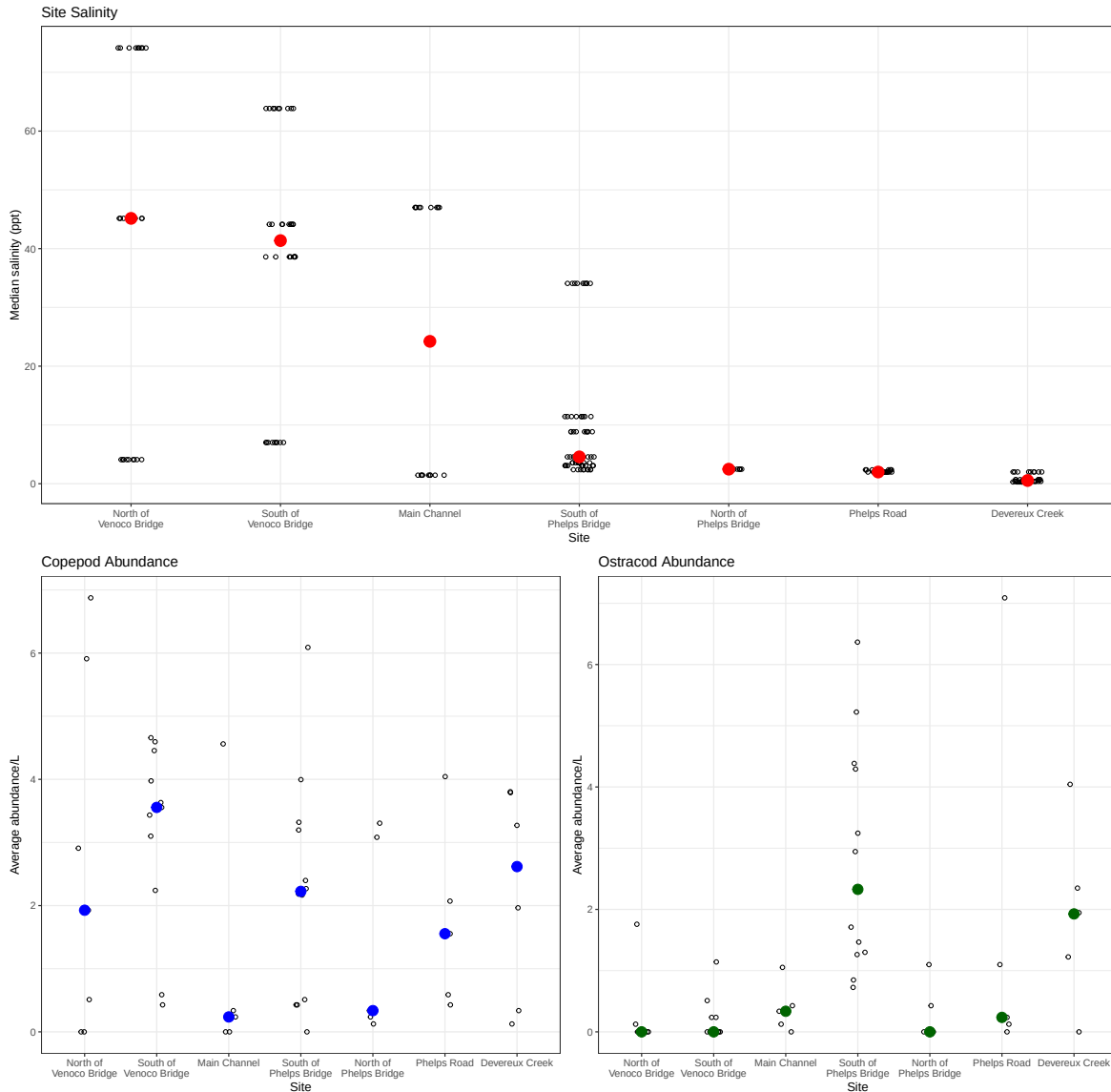
```

ostracods_plot <- ggplot(data = ost_abundance,
                        # x-axis: site
                        # y-axis: mean abundance/L
                        mapping = aes(x = site_full,
                                      y = log_ave_lit)) +
# first layer: points to represent observations
geom_jitter(height = 0,
            shape = 21,
            width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
            fun = "median",
            size = 4,
            color = "darkgreen") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "Average abundance/L",
     title = "Ostracod Abundance") +
# different theme
theme_bw()

# arranging plots
(salinity_plot) / (copepods_plot + ostracods_plot) +
plot_annotation(title = "Comparing Salinity, Copepod Abundance, and
↪ Ostracod Abundance across sites at NCOS")

```

Comparing Salinity, Copepod Abundance, and Ostracod Abundance across sites at NCOS



e. Interpret your figure

South of Phelps Bridge and Devereux Creek had the highest Average abundance/L measurements across sites for Ostracods. The other sites had pretty consistent lower average abundance/L measurements. North of Venoco Bridge and South of Phelps Bridge had the highest Average abundance/L measurements across sites for Copepods. Main Channel and North of Phelps Bridge had the lowest average abundance/L measurements for Copepods. I could determine

these comparisons by observing how high up on the y axis data points lie, and compare across sites on the x axis.

There does not appear to be a strong correlation between site salinity and ostracod abundance. The site with the highest average salinity (ppt) measurements, South of Vencoco Bridge, does not have a notably different average abundance/L measurements than the other sites. Similarly, the site with the highest average abundance/L measurements, South of Phelps Bridge, did not have comparably high of low salinity (ppt) points along the y axis.

2. Visualizing total abundance as a function of salinity

a. Plan your figure

- x-axis: med_sal
- y-axis: log_ave_lit
- color: taxon
- geometry: geom_point()
- facets: facet_wrap(~ taxon, scales = "free")

b. Wrangle your data

```
# create new object from aq_ins_clean
abundance_salinity <- aq_ins_clean |>
  # create log-transformed abundance column
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # remove underscores from taxon names
  mutate(taxon = str_to_sentence(str_replace_all(taxon, "_", " "))) |>
  # group by taxon
  group_by(taxon) |>
  # calculate max abundance
  mutate(max_abundance = max(log_ave_lit, na.rm = TRUE)) |>
  # un group data frame
  ungroup() |>
  # reorder taxa by maximum abundance
  mutate(taxon = fct_reorder(taxon,
                             max_abundance,
                             .desc = TRUE)) |>

# keep needed columns
select(taxon,
       med_sal,
       log_ave_lit)
```

```
# display 10 random rows
abundance_salinity |>
  slice_sample(n = 10)
```

```
# A tibble: 10 x 3
  taxon          med_sal log_ave_lit
  <fct>          <dbl>     <dbl>
1 Annelida polychaete    NA         0
2 Cladocera              NA         0
3 Copepod                NA        4.66
4 Cladocera            38.6         0
5 Cladocera             8.84        0.236
6 Ostracod              NA        0.125
7 Nematode              NA         0
8 Nematode             8.84         0
9 Annelida polychaete    4.1         0
10 Copepod              NA        0.336
```

c. Code your figure

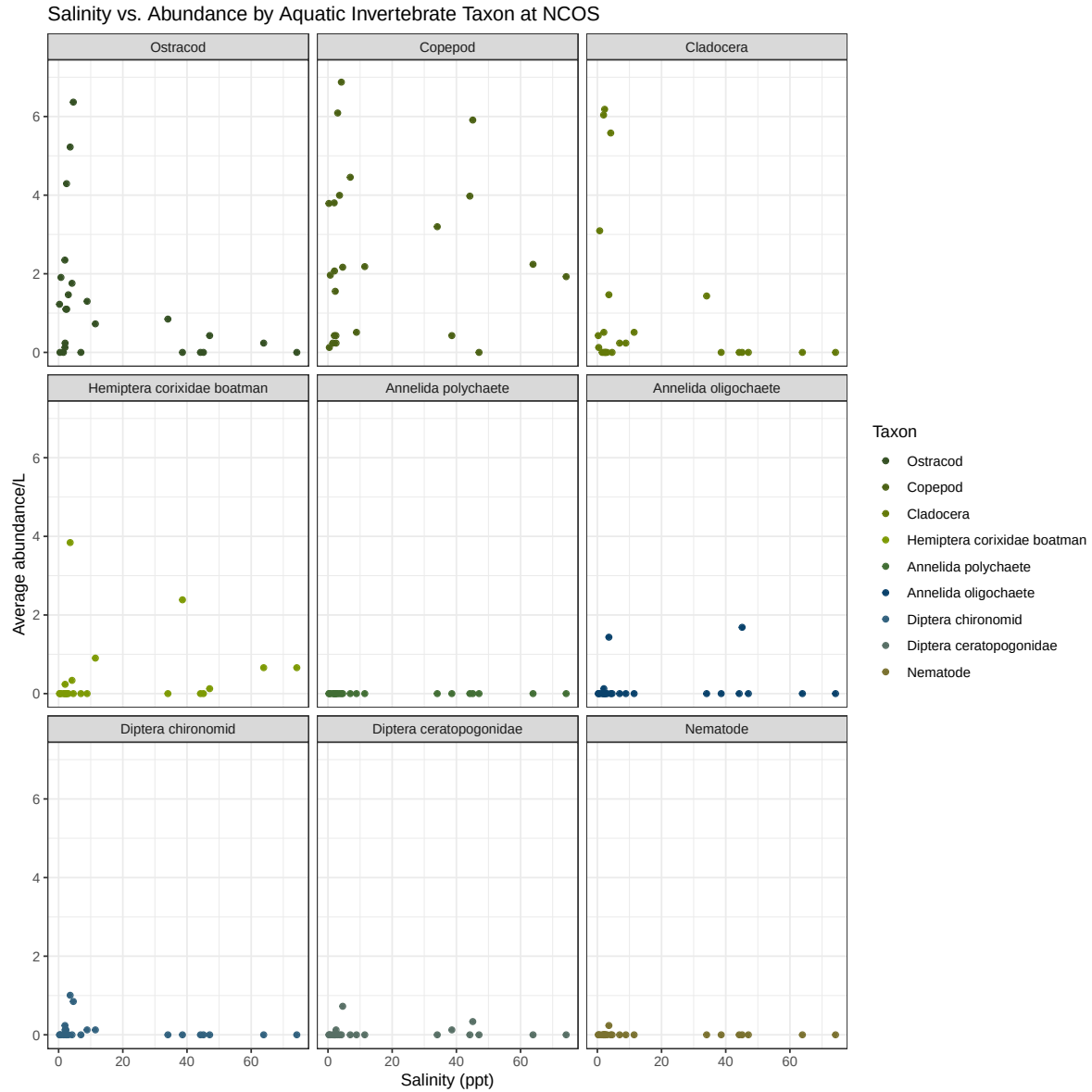
```
# base layer
abu_sal_plot <- ggplot(data = abundance_salinity,
  # x-axis: salinity
  # y-axis: mean abundance/L
  mapping = aes(x = med_sal,
    y = log_ave_lit,
    color = taxon)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
    shape = 21,
    width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
    fun = "median",
    size = 1) +
  facet_wrap(~ taxon) +
  # re labelling x- and y-axis, adding title
  labs(x = "Salinity (ppt)",
    y = "Average abundance/L",
```

```

    color = "Taxon",
    title = "Salinity vs. Abundance by Aquatic Invertebrate Taxon at
    ↪ NCOS") +
# change color palette, indicate 9 colors for 9 taxon groups
scale_color_manual(values = natparks.pals("Everglades", n = 9)) +
# different theme
theme_bw()

abu_sal_plot

```



d. Interpret your figure

Ostracods and Cladocera exist in the highest average abundance at lower salinities, between 0 and 10 ppt. Annelida polychaete, Annelida oligochaete, Diptera chironomid, Diptera ceratopogonidae and Nematodes all exist in low average abundance, typically less than 1/L, across the entire salinity range, 0-80 ppt. I could determine both of these patterns by observing where the highest concentration of data points lies along both the y and x axes. Copepods and

Hemiptera corixidae boatman are a bit more variable; there are data points high and low along both the average abundance and salinity axes.